Naval Architecture

Seakeeping and Motion Analysis of the Richelieu

Students :
Marie Bernard-Lameau
Anaëlle Havret
Lisa Jabot

Teacher :
Etienne Dufour

19 January 2026

Table of Contents

1 Introduction

The Richelieu was a French battleship commissioned in 1940, representing one of the most advanced warships of her time. Designed in a context of growing international tensions, she operated in multiple theaters around the world during the Second World War, illustrating the global reach and strategic importance of naval power in the mid-twentieth century. This report focuses on the naval architectural aspects of the Richelieu. The work is structured in three main parts : first, the characterization of the sea state through an environmental assessment ; second, the analysis of the ship's stability ; and finally, the study of the vessel's dynamic response to wave excitation.

2 Environmental assessment

This section presents the environmental assessment of the sea state. The objective is to characterize the wave field using the JONSWAP spectrum and the associated reconstructed time series.

The total number of observation is 99,999. Thus, 90^{th} percentile of $Hs = 0.9 \cdot 99,999 = 89,999.1$ observations of Hs less than the researched value. If we add the observations from $Hs = 0.5m$ to $Hs = 4.5m$, we obtain 81,852 ($< 89,999.1$) observations, and from $Hs = 0.5m$ to Hs=5.5m, we have 90,180 observations ($> 89,999.1$). Thus, we chose $Hs = 5.5m$ as the maximum 90^th percentile. The most propable Tz associated to this value is $Tz = 9.5s$ ($\mathrm{P}_{(Hs,Tz)} = 2,372.7$).

According to the DNV-RP-C205 (3.5.5.4), we can compute the wave peak period T_p :

$$T_p = T_z(\gamma + \gamma^2 + \gamma^3) \tag{1}$$

For $\gamma = 3.3$ (normal sea conditions) :

$$T_p = 1.2859 \cdot T_z \tag{2}$$

Thus, $T_p = 12.2s$.

We can plot the JONSWAP spectrum using the following formula for frequencies ranging from $0.01Hz$ to $1.15Hz$ with a resolution of $0.01Hz$. The resulting spectrum shows a peak frequency of about $0.6rad/s$, which corresponds to a period of approximately $10.5s$.

$$S_j(\omega) = A_\gamma . \frac{5}{16} . H_s^2 . \frac{\omega_p^4}{\omega^5} . exp(-\frac{5}{4}(\frac{\omega}{\omega_p})^{-4}) . \gamma^{e^{(-0.5(\frac{\omega-\omega_p}{\sigma.\omega_p})^2)}} \tag{3}$$

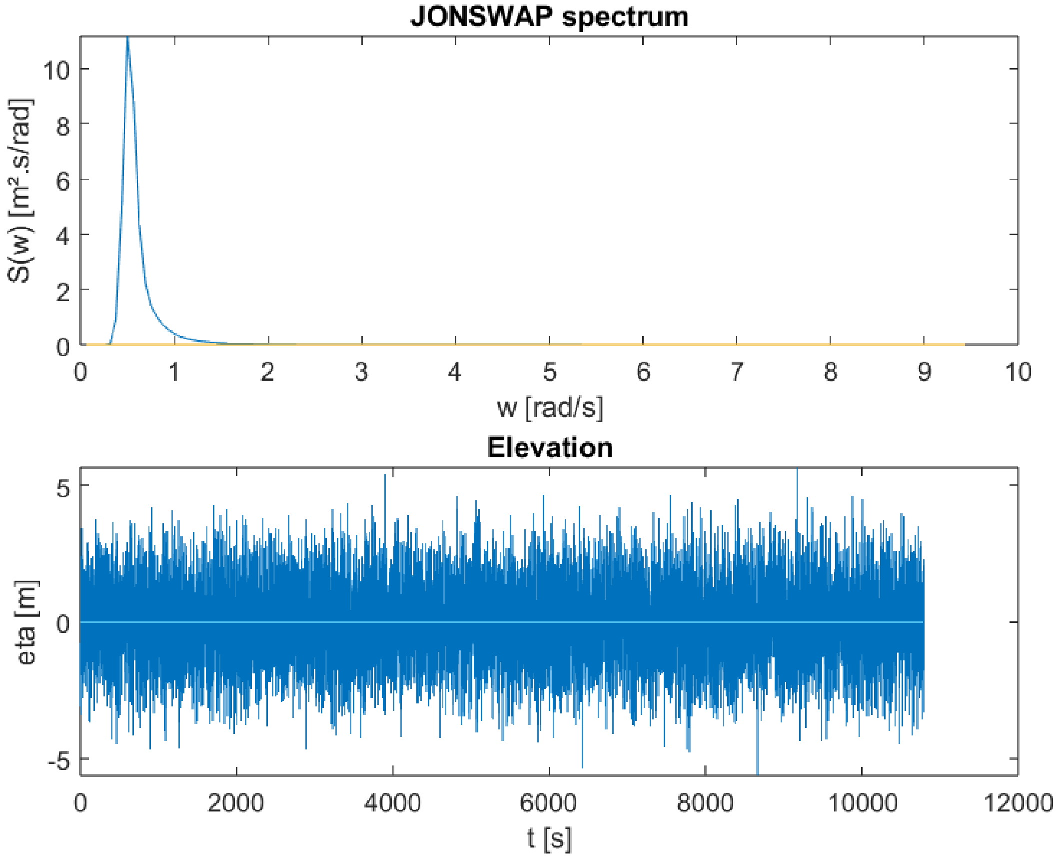

Figure 1: JONSWAP spectrum and elevation of the wave

The wave elevation time series over a duration of one hour is reconstructed from the JONSWAP spectrum by superposing the contributions of 100 discrete frequency components. Each component is associated with an amplitude derived from the spectral density and a random phase uniformly distributed over [0,2π].

$$\eta(t) = \sum_{k=0}^{100} \sqrt{2\, S_J(w_k)\, \Delta w_k}\, \cos(2\pi w_k t + \phi_k)\,, \tag{4}$$

With $\phi_k \sim \mathcal{U}(0, 2\pi)$, $\quad \Delta w_k = \frac{w_{\max} - w_{\min}}{100}$.

The obtained surface elevation curve η, expressed in meters as a function of time, clearly corresponds to a one-hour period, with waves reaching heights of up to $5m$.

The frequency range [0.0Hz,1,5Hz], so a period range [0.66s,100s] was selected to capture all physically relevant gravity-wave components described by the JONSWAP spectrum. Periods longer than 100s correspond to negligible low-frequency energy, while periods shorter than 0.66s are mainly associated with capillary waves, which are not represented by the JONSWAP formulation. This range therefore ensures an accurate reconstruction of the wave elevation time series while maintaining physical relevance.

3 Ship Mechanical and Physical Properties

In this section, we consider the origin of the reference frame amidship, centerline, and keel.

The main goal of the section is to assess the stability and roll period of the ship.

3.1 Center of Gravity and Inertia Matrix

The equilibrium of the ship depends on the cargo and its weight distribution.

First, we identified the tanks containing cargo and consumables, as their positions affect the overall center of gravity of the ship. The analysis included 7 ballast tanks, 13 mazout tanks, 2 oil tanks, and 2 water tanks.

Based on the plans and the graduated frame, we determined the length of each tank. Assuming each tank is a cuboid, we calculated the position of their longitudinal centers of gravity (LCG), located at the midpoint of their length. Using a ruler and known dimensions, we computed the scale of the plan by dividing the length of the drawing in centimeters by the real length in meters. Then, we obtained the height and the vertical center of gravity (VCG) of each tank.

The volume of the tanks being given, we divided the volume by the length multiplied by the height to find the width. We verified that the values were plausible (less than the beam of the ship). No information was provided regarding the transversal centers of gravity (TCG), so we assumed all tanks are centered, giving $TCG = 0m$.

To assess the mass of the tanks, we used the following densities :

$$\rho_{\text{water}} = 1 \text{ te/m}^3, \quad \rho_{\text{oil}} = 0.875 \text{ te/m}^3, \quad \rho_{\text{mazout}} = 0.850 \text{ te/m}^3.$$

Knowing the volume and density, we obtained the mass by multiplication of these quantities.

We also determined the basic metrics of the hull, including its length, width, height, and mass. The dimensions were determined using the plans, yielding : length $= 247.885m$, beam $= 33m$, and depth $= 18.27m$. No mass information was available. The most relevant value found was a standard displacement of 37,850 tons on this website, which we adopted as the lightship displacement.

The Excel file provided was filled with these values, allowing us to compute the displacement, the inertial matrix, the CoG, and the FSM-corrected CoG of the ship.

The total displacement of the ship is $39,555.84te$, which falls within the expected range of $36,243te$ to $44,297te$. The inertial matrix is :

$$\mathbf{J} = \begin{bmatrix} 5.18 \times 10^6 & 0 & 0 \\ 0 & 2.05 \times 10^8 & 0 \\ 0 & 0 & 2.08 \times 10^8 \end{bmatrix} \quad (\text{te}\cdot\text{m}^2).$$

In standard conditions, the center of gravity is located at the following distances from the origin of the chosen reference frame:

$$\mathbf{CoG}^{(\text{corrected})} = \begin{pmatrix} \text{LCG} = 123.71 \text{ m} \\ \text{TCG} = 15.43 \text{ m} \\ \text{VCG} = 8.84 \text{ m} \end{pmatrix}.$$

3.2 Transverse Metacentric Radius (BM)

Let's compute the transverse metacentric radius. According to the lecture notes :

$$BM = \frac{I_{WPA}}{V_{ship}} \tag{5}$$

with,

- I_{WPA} the inertia of the Water Plane Area (the surface of the boat at the waterline level) in $[m^4]$,
- and V_{ship} the volume of the ship below the waterline in $[m^4]$.

Let's first compute I_{WPA}. As the hull is curved, we use the following formula :

$$I_{WPA} = \frac{2}{3}\int_0^{L_{pp}} b^3(x)dx \tag{6}$$

with,

- $b(x)$ the half breadth of the boat in [m],
- L_{pp} the length between perpendiculars in [m].

According to the body plan given, we have the half breadth of the boat every 12.1 m along the centerline, at the waterline level. We converted the given values to meters. Dividing the boat into 20 sections along the centerline, we approximate $b(x)$ by a step function. Thus, we can discretize the integral into a finite sum :

$$I_{WPA} = \frac{2}{3}\sum_1^{20}\int_{i*d_s}^{(i+1)*d_s} b_i^3 \tag{7}$$

with,

- d_s the distance between sections,
- and b_i the half breadth of the i^{th} section.

The remaining integral becomes the area of a rectangle with width d_s and height b_i^3. That is to say, d_s multiplicated by b_i^3.

$$I_{WPA} = \frac{2}{3}\sum_0^{20} d_s b_i^3 \tag{8}$$

Using this formula, we obtain $I_{WPA} = 326{,}437\ [m^4]$.

Let's now compute the volume of the boat below the waterline. We consider that the boat is not moving along the $z-axis$ compared to the water surface. That is to say, the forces exerted on the boat along this axis compensate each other.

$$||\vec{B}|| = ||\vec{W}|| \tag{9}$$

- $\vec{B}$, the buoyancy,
- $\vec{W}$, the weight of the boat.

Thus,

$$\rho_{water}\ g\ V_{ship} = g\ m_{ship} \tag{10}$$

$$V_{ship} = \frac{m_{ship}}{\rho_{water}} = 39,288\ m^3 \tag{11}$$

- V_{ship}, the volume of the ship below the waterline in $[m^3]$,
- ρ_{water}, the density of seawater (here, 1025 kg/m^3),
- m_{ship}, the displacement of the ship in [kg],$40,270.10^3$.

Knowing both V_{ship} and I_{WPA}, we can now compute BM :

$$BM = \frac{I_{WPA}}{V_{ship}} = 8.31\ m \tag{12}$$

3.3 Metacentric height (GM)

It has been demonstrated in class that :

$$GM = KB + BM - KG \tag{13}$$

With :

- BM, the transverse metacentric radius, here $8.31m$
- KG, the elevation of the center of gravity. Its value is equal to the VCG previously computed, is equal to $8.84m$
- KB, the elevation of the center of buoyancy, which corresponds to the VCB given in the instructions is equal to $5.11m$

Thus, we have $GM = 4.58\ m$. As this value is positive, the righting arm and the righting moment will tend to bring back the ship in its equilibrium position. The boat is stable.

3.4 Ship Roll Period

According to the lecture notes,

$$T_r = 2\pi\sqrt{\frac{I_x}{m_{ship}.g.GM}} \tag{14}$$

Using the inertia matrix previously found, $I_x = 5.18 \cdot 10^6\ te \cdot m^2$, and we know that $m_{ship} = 40,270\ te$, $GM = 4.58\ m$, and $g = 9.81\ m \cdot s^{-2}$. Thus, $T_r = 10.6$ s. This value is unexpectedly high considering the significant GM value, but it results from the very large roll inertia ($I_x = 5.18 \times 10^6$ te·m^2). This roll period is high enough to ensure comfort for passengers.

4 Dynamic Assessment

This section investigates the ship's dynamic response through the analysis of Response Amplitude Operators (RAOs), linking the vessel motions to the sea states previously characterized in Part 1.

4.1 Ship response

Based on the ship mechanical analysis, we obtain the Response Amplitude Operators (RAOs) for the 6 degrees of freedom (surge, sway, heave, roll, pitch, yaw). RAOs are defined as follows :

$$RAO_i(\omega) = \frac{X^F_{(i)}(\omega)}{\eta(\omega)} \tag{15}$$

with $X^F_{(i)}$ the motion of the CoG of the boat in *radians* (pitch, roll, yaw) or *meters* (heave, sway, surge). They allow us to quantify the motion of the CoG of the boat due to the waves.

The text files contain the RAOs values for front waves (angle = 180°), quartering waves (angle = 135°), and beam waves (angle = 90°). They are organised as follows :

- 1^{st} column : angular frequencies between $0.01 rad/s$ and $2 rad/s$,
- 2^{nd} to $4th$ columns : modules of the RAOs associated to beam, quartering and front waves,
- 5^{th} to $7th$ columns : arguments of the RAOs associated with beam, quartering and front waves.

Thanks to a matlab script, we extract the data and create vectors representing each column. We can now plot the modules of the RAOs in terms of the angular frequency (Figure 2). Each plot represents a DoF and contains the data for the 3 wave angles.

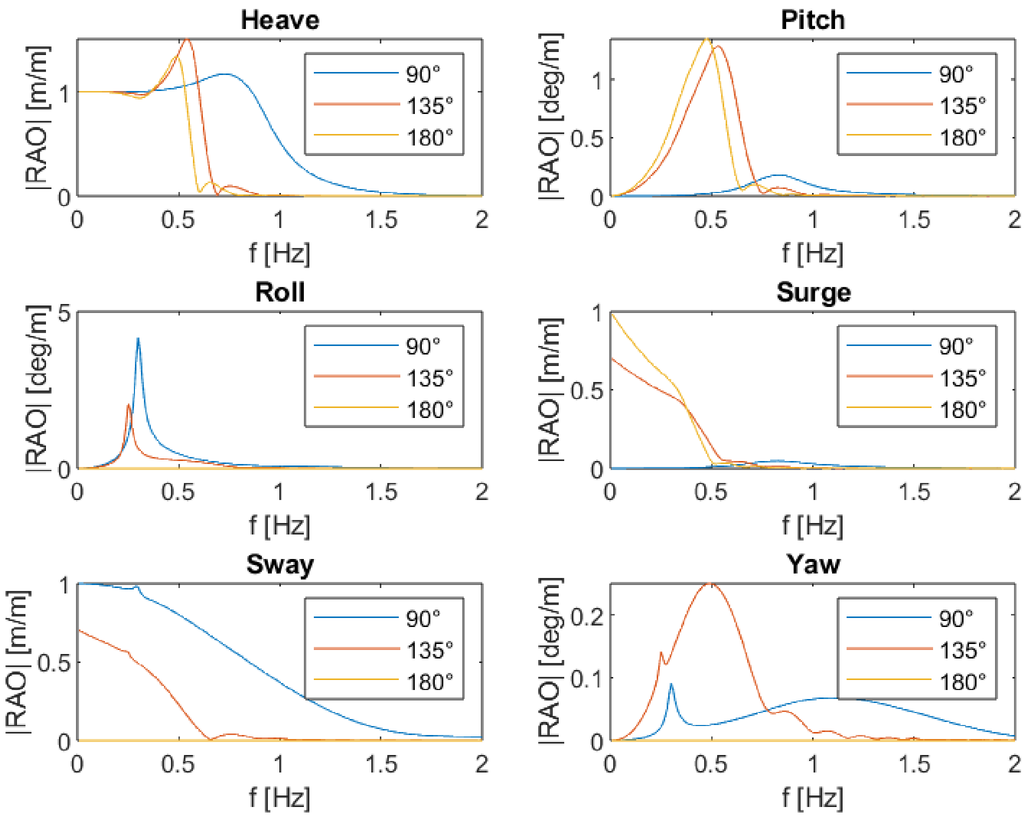

Figure 2: 6DoF RAO

Using the RAO spectrum together with the sea state, we can compute and plot the dynamic response of the Richelieu using the following formula :

$$x(t) = |RAO(w)|\eta(t)cos(wt + arg(RAO)) \tag{16}$$

As each RAO is defined at specific pulsation ω that differ from those used for the wave elevation spectrum, to obtain the ship response at the desired pulsation (often requiring finer resolution) interpolation is therefore necessary.

For a beam wave, Figure 3 shows that the surge and the pitch response remain very low, which is consistent with the expected behaviour. Indeed, the boat cannot significantly move forward in the direction of its axis or turn along the 90° axis but it can move depending on the other degrees of freedom.

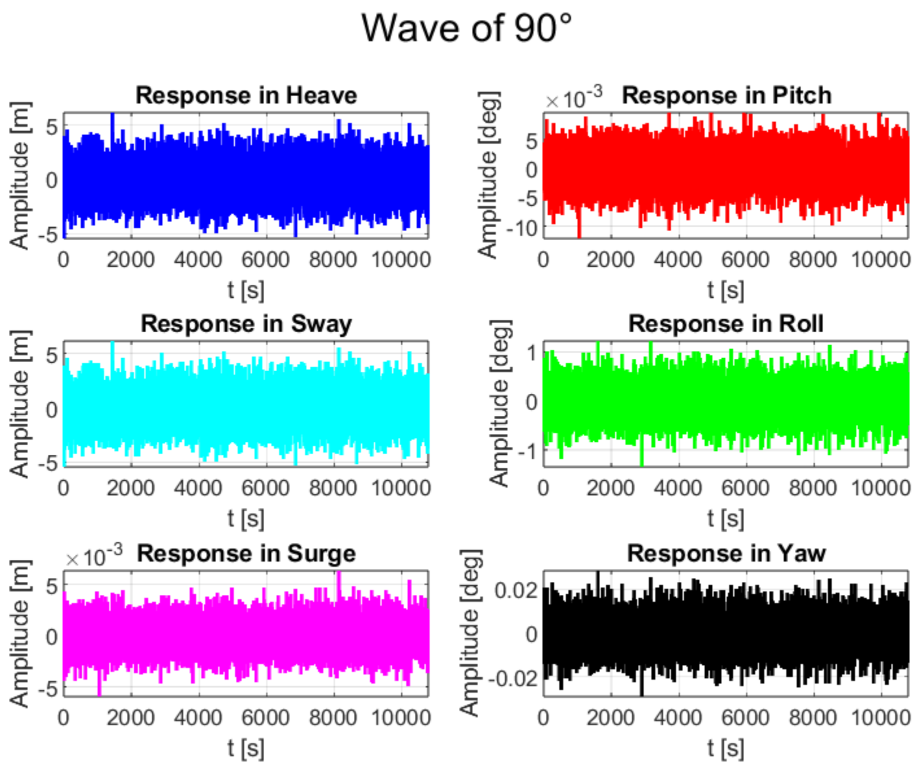

Figure 3: Ship response for a beam wave

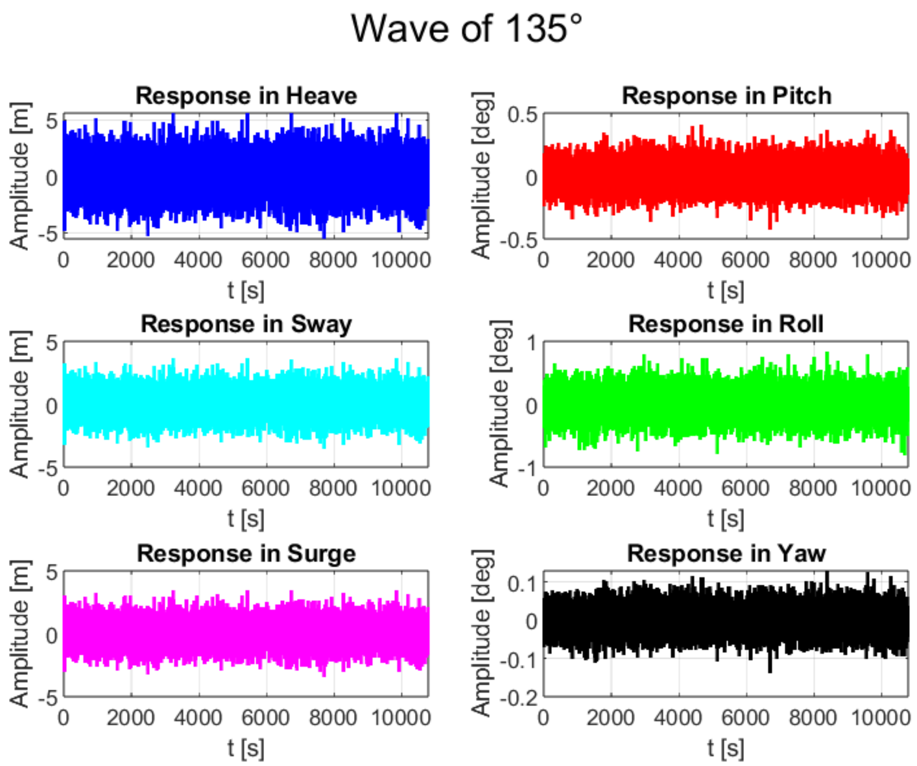

Figure 4: Ship response for a quartering wave

On the other hand, for a quartering wave, the boat can move in all directions, which is consistent with the graphs obtained in Figure 4.

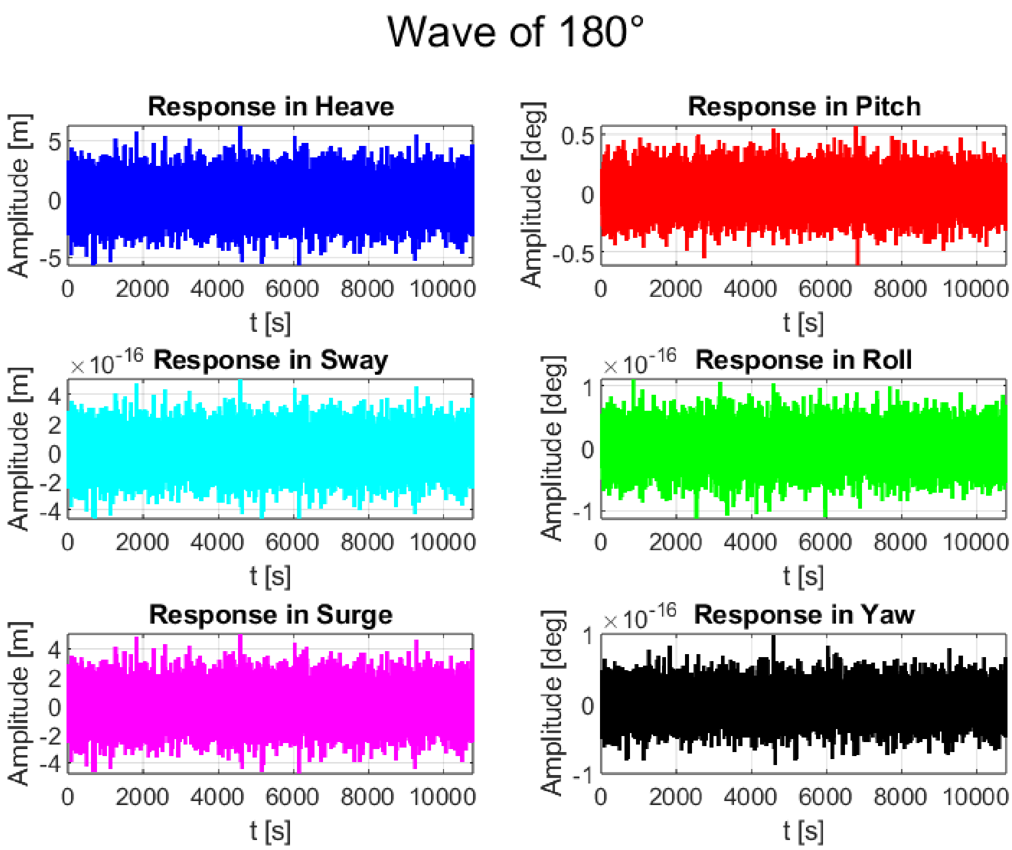

Figure 5: Ship response for a front wave

Finally, for a front wave the ship has 3 degrees of freedom : heave, pitch and surge (Figure 5) which is also consistent with the expected behaviour.

4.2 Ship spectral response

The ship spectral response is defined as follows :

$$S_R(\omega) = RAO^2(\omega) \cdot S_W(\omega) \tag{17}$$

With $S_W(\omega)$ the incoming wave spectrum in $[m^2/Hz]$.

We can therefore plot the response spectrum of the Richelieu for the 6 degrees of freedom :

Figure 6 shows that for a beam wave, the roll, heave and sway spectrum are significant because the ship is excited laterally. This is therefore coherent with the ship response obtained in Figure 3. Pitch and sway movements are negligible, while yaw is low.

For a quartering wave (Figure 7), only the spectrum in yaw seems low while the others are significant.

Finally for a front wave, Figure 8 shows that the same 3 degrees of freedom as the ship response in Figure 5 are significant : heave, pitch and surge.

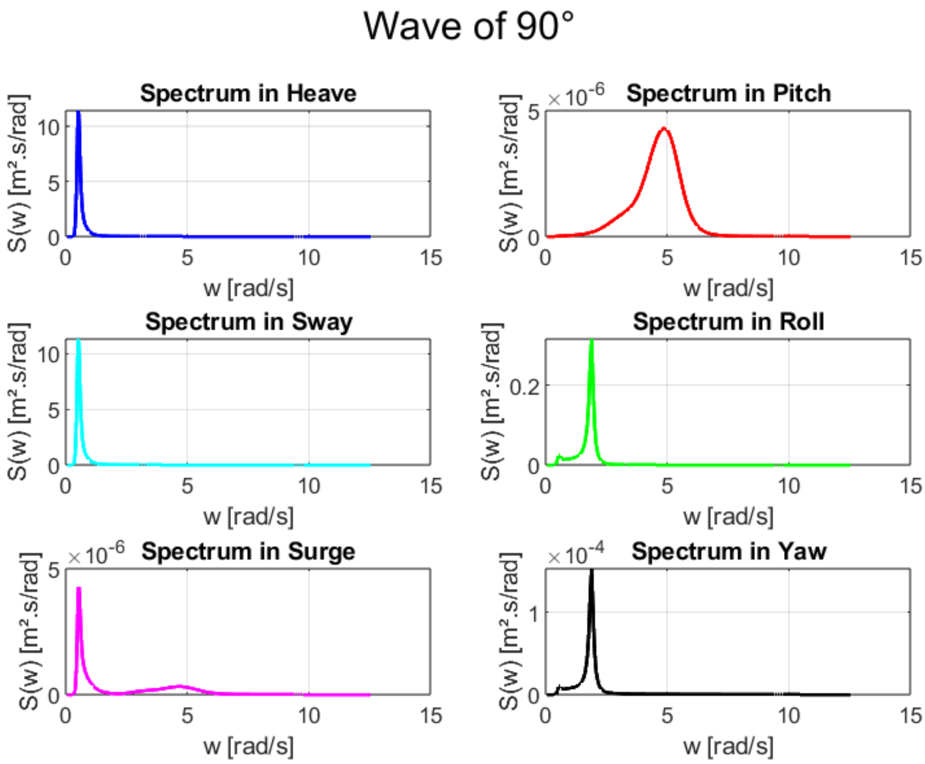

Figure 6: Ship spectral response for a beam wave

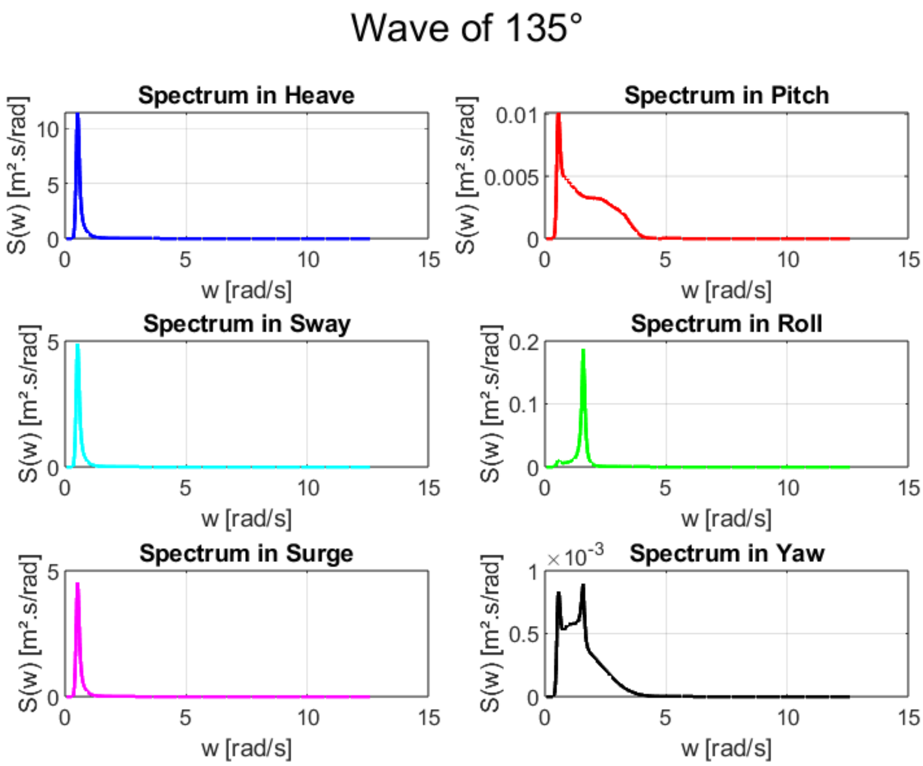

Figure 7: Ship spectral response for a quartering wave

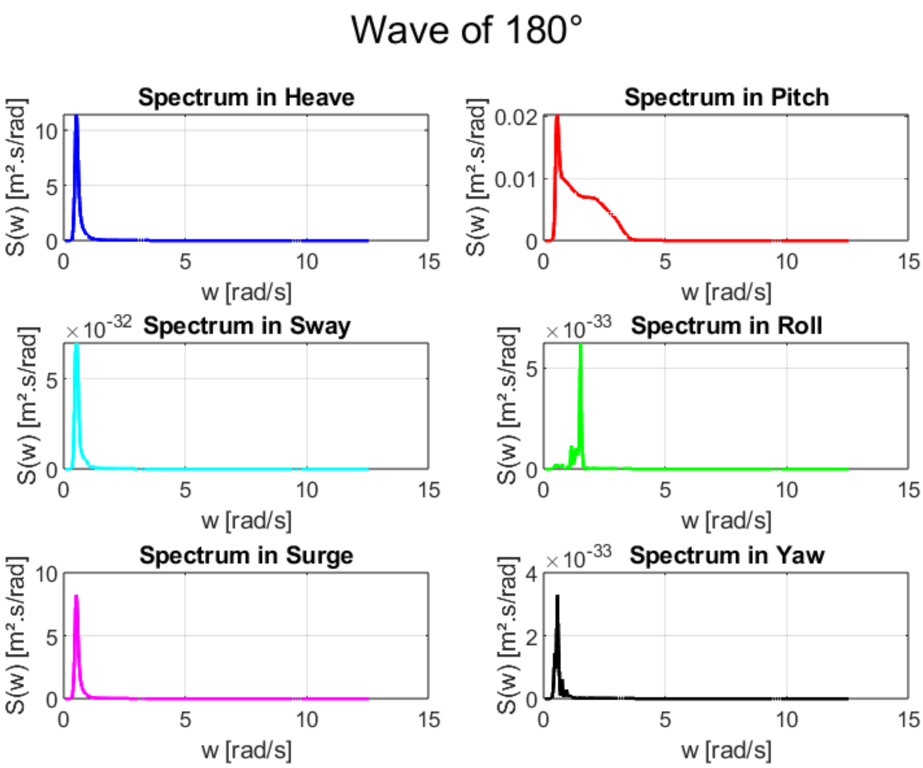

Figure 8: Ship spectral response for a front wave

4.3 Ship response analysis

The significant response of the ship is calculated using the 0^{th} order moment of the response spectrum, which can be calculated as :

$$x_r = 4.\sqrt{m_0} \tag{18}$$

with,

$$m_k = \int_{\omega=0}^{\infty} \omega^k \cdot RAO^2(\omega) \cdot S_W(\omega) d\omega = \int_{\omega=0}^{\infty} \omega^k S_R(\omega) d\omega \tag{19}$$

thus,

$$m_0 = \int_{\omega=0}^{\infty} S_R(\omega) d\omega \tag{20}$$

Then, we can compute the zero-up crossing period of the response T_z as :

$$T_z = 2\pi\sqrt{\frac{m_0}{m_2}} \tag{21}$$

Finally, by assuming that the maximum of the ship response follows a Gumbel distribution, we can obtain the most probable maximum of x_R using the following formula :

$$x_{MPM} = \sqrt{2 \cdot m_0 \ln(\frac{T}{T_z})} \tag{22}$$

with T being the length of the observation in time.

We finally obtain the following values for 3 specific degrees of freedom :

		90°	135°	180°
Ship significant response (xr)	Heave [m]	6.17	6.17	6.17
	Pitch [deg]	0.01	0.42	0.57
	Roll [deg]	1.30	0.86	0.00
Zero up-crossing period (Tz)	Heave [s]	9.83	9.85	9.90
	Pitch [s]	1.34	3.23	3.62
	Roll [s]	3.57	4.14	4.46
Most probable maximum amplitude (xmpm)	Heave [m]	5.77	5.77	5.77
	Pitch [deg]	0.01	0.43	0.57
	Roll [deg]	1.31	0.85	0.00

Figure 9: Table of remarkable values for the boat's roll, pitch and heave response for three waves with different incident angles.

4.3.1 Practical application

Now, let's consider that the main gun of the Richelieu have been rotated from 90° relative to the ship's longitudinal axis. Then, it is elevated by an angle of 10°. A wave approaches at 90°, and we want to determine the variation of the gun's angle. Therefore, we need the roll angle of the ship for a wave coming from 90°.

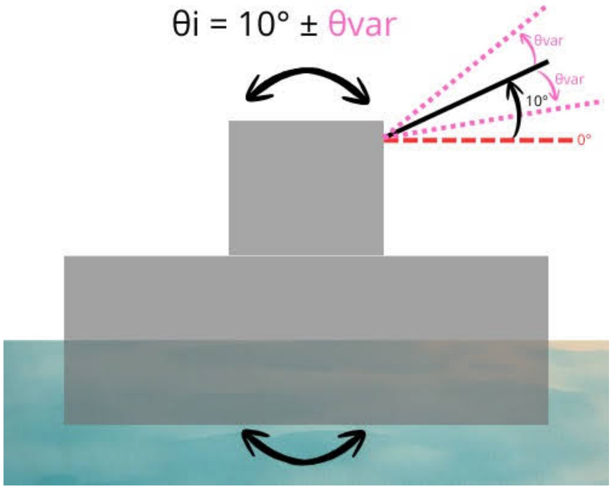

Figure 10: Ship response in roll for a beam wave

According to the table on Figure 9, this angle is the most probable maximum amplitude that is equal to 1.31°.

Now that the main gun is ready to fire, we want to determine its firing range as the boat's angle of inclination varies, knowing that the shell is launched at a velocity of $830m/s$. This is a classic projectile motion problem. We calculate the flight time and the horizontal range of the shell under gravity only, using the initial velocity components in the x and y directions.

By integrating the accelerations twice, we obtain the following equations of motion :

$$\begin{cases} x(t) = v_{0_x} t \\ y(t) = y_0 + v_{0_y} t - \dfrac{1}{2} g t^2 \end{cases}$$

Here, we have $v_0 = 830\ m/s$ and :

$$\vec{v_0} = v_0 \cos\theta\, \vec{\mathbf{e}_x} + v_0 \sin\theta\, \vec{\mathbf{e}_y} = v_{0_x} \vec{\mathbf{e}_x} + v_{0_y} \vec{\mathbf{e}_y} \tag{23}$$

Thanks to the Richelieu's plans, we also have: $y_0 = 13.25m$, the height between the waterline and the gun axis.

We can find the flight time t_1 when $y(t_1) = 0$, which corresponds to the moment when the shell hits the water:

$$t_1 = \frac{v_0 \sin\theta + \sqrt{(v_0 \sin\theta)^2 + 2gy_0}}{g} \tag{24}$$

Finally, the horizontal distance (range) of the shot is obtained as:

$$x(t_1) = v_0 \cos\theta \, t_1 = \frac{v_0 \cos\theta \left(v_0 \sin\theta + \sqrt{(v_0 \sin\theta)^2 + 2gy_0}\right)}{g} \tag{25}$$

θ_{min} $[rad]$	0.15	P_{min} $[m]$	21,086
θ_i $[rad]$	0.17	P_{target} $[m]$	24,093
θ_{max} $[rad]$	0.19	P_{max} $[m]$	27,053

Table 1: Variation of the range of the shot

With $P = x(t_1)$ in meters, the range of the shot for different angles.

The firing angle can vary between 8.7° and 11.3°, so the range of the shot is between $21km$ and $24km$. Therefore, the shell's accuracy is reduced, as it may not hit its target.

However, the range of the shot seems very long, which indicates an error in calculation or reasoning.

Appendices

For the purpose of simplicity, we have removed the plots from the programmes in order to focus on how to obtain the curves.

We also only show the codes for a beam wave because those for a front and quartering wave are identical, with only the RAO columns that we extract from the files provided needing to be adapted.

```
clear all
clc
close all

elevation=fopen('C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\elevation.txt',"w");
fprintf(elevation,"t [s] \t eta [m] \n");

% Data :
Hs=5.5;
Tp=12.22;
wp=2*pi/Tp;
gamma=3.3;
Ag=0.2/(0.065*gamma^0.803+0.135);
f=0.01:0.01:1.5;
w=2*pi*f;

% JONSWAP :
S=zeros(length(w));
for i=1:length(w)
    if w(i)<wp
        sigma=0.07;
    elseif w(i)>wp
        sigma=0.09;
    end
S(i)=Ag*5/16*Hs^2*wp^4/w(i)^5*exp(-5/4*(w(i)/wp)^(-4))*gamma^exp(-0.5*((
    w(i)-wp)/sigma)^2);
end

N=150;
t=1:1:3*3600;
eta=zeros(length(t));
dw=w(2)-w(1);

for i=1:length(t)
    for j=1:N
        phi=-pi+rand(1)*2*pi; % random phase
        eta(i)=eta(i)+sqrt(2*dw*S(j))*cos(w(j)*t(i)+phi);
    end
    fprintf(elevation,"%f \t %f \n",t(i),eta(i));
end

fclose(elevation);
```

Listing 1: Environment

This algorithm simulates ocean wave elevation over time using the JONSWAP wave spectrum model (*see Section 2*). It generates a realistic 3-hour wave time series based on input parameters : $Hs = 5.5m$, $Tp = 12.22s$ and $\gamma = 3.3$.

```
function[S]=JONSWAP(w1)
S=zeros(length(w1));
Hs=5.5;
Tp=12.22;
wp=2*pi/Tp;
gamma=3.3;
Ag=0.2/(0.065*gamma^0.803+0.135);
for i=1:length(w1)
    if w1(i)<wp
        sigma=0.07;
    elseif w1(i)>wp
        sigma=0.09;
    end
S(i)=Ag*5/16*Hs^2*wp^4/w1(i)^5*exp(-5/4*(w1(i)/wp)^(-4))*gamma^exp
    (-0.5*((w1(i)-wp)/sigma)^2);
end
end
```

Listing 2: JONSWAP function

This algorithm is a function that calculates the JONSWAP spectrum for a range of pulsation ω. We use it for our other programmes *(see below)*.

```
clear all
clc
close all

%% Dynamics :

RAO_heave=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_heave.txt");
RAO_pitch=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_pitch.txt");
RAO_roll=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_roll.txt");
RAO_surge=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_surge.txt");
RAO_sway=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_sway.txt");
RAO_yaw=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_yaw.txt");

f=RAO_heave(:,1);

%% Ship response :

elevation=load('C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\elevation.txt');
t=elevation(:,1);
fj=0.01:0.001:2;
wj=2*pi*fj;
```

```
% Interpolation by a linear function
RAO_heave_interp = interp1(f, RAO_heave(:,2:end), fj, 'linear');
RAO_pitch_interp = interp1(f, RAO_pitch(:,2:end), fj, 'linear');
RAO_roll_interp = interp1(f, RAO_roll(:,2:end), fj, 'linear');
RAO_surge_interp = interp1(f, RAO_surge(:,2:end), fj, 'linear');
RAO_sway_interp = interp1(f, RAO_sway(:,2:end), fj, 'linear');
RAO_yaw_interp = interp1(f, RAO_yaw(:,2:end), fj, 'linear');

x_heave = zeros(length(t), 1);
x_pitch = zeros(length(t), 1);
x_roll = zeros(length(t), 1);
x_surge = zeros(length(t), 1);
x_sway = zeros(length(t), 1);
x_yaw = zeros(length(t), 1);

dwj=wj(2)-wj(1);

for i=1:length(t)
for k = 1:length(wj)

    %RAOs at this frequency (first column = 90  )
    RAO_h = RAO_heave_interp(k, 1);
    RAO_p = RAO_pitch_interp(k, 1);
    RAO_r = RAO_roll_interp(k, 1);
    RAO_sur = RAO_surge_interp(k, 1);
    RAO_sw = RAO_sway_interp(k, 1);
    RAO_y = RAO_yaw_interp(k, 1);

    ARAO_h = RAO_heave_interp(k, 4);
    ARAO_p = RAO_pitch_interp(k, 4);
    ARAO_r = RAO_roll_interp(k, 4);
    ARAO_sur = RAO_surge_interp(k, 4);
    ARAO_sw = RAO_sway_interp(k, 4);
    ARAO_y = RAO_yaw_interp(k, 4);

    phi=-pi+rand(1)*2*pi;

    %Contribution to the time response
    x_heave(i) = x_heave(i) + sqrt(2*dwj*JONSWAP(wj(k))) .* RAO_h .* cos
   (wj(k)*t(i) + phi + angle(ARAO_h));
    x_pitch(i) = x_pitch(i) + sqrt(2*dwj*JONSWAP(wj(k))) .* RAO_p .* cos
   (wj(k)*t(i) + phi + angle(ARAO_p));
    x_roll(i) = x_roll(i) + sqrt(2*dwj*JONSWAP(wj(k))) .* RAO_r .* cos(
   wj(k)*t(i) + phi + angle(ARAO_r));
    x_surge(i) = x_surge(i) + sqrt(2*dwj*JONSWAP(wj(k))) .* RAO_sur .*
   cos(wj(k)*t(i) + phi + angle(ARAO_sur));
    x_sway(i) = x_sway(i) + sqrt(2*dwj*JONSWAP(wj(k))) .* RAO_sw .* cos(
   wj(k)*t(i) + phi + angle(ARAO_sw));
    x_yaw(i) = x_yaw(i) + sqrt(2*dwj*JONSWAP(wj(k))) .* RAO_y .* cos(wj(
   k)*t(i) + phi + angle(ARAO_y));
end
end
```

Listing 3: Dynamic : 90°

This code simulates ship motion responses in six degrees of freedom using RAO data. It loads pre-computed RAOs for heave, pitch, roll, surge, sway, and yaw, visualizes their frequency response, then combines them with a JONSWAP wave spectrum to generate realistic time-domain ship motions (*see Section 4.1*).

The simulation shows how the ship moves over time when encountering waves from a 90° direction here.

```
clear all
clc
close all

fprintf('\n==== Beam wave (90  ) ====\n');

RAO_heave=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_heave.txt");
RAO_pitch=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_pitch.txt");
RAO_roll=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_roll.txt");
RAO_surge=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_surge.txt");
RAO_sway=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_sway.txt");
RAO_yaw=load("C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\RAO\cog_yaw.txt");

f=RAO_heave(:,1);
elevation=load('C:\Users\lisa\Documents\SeaTech\2A\Naval Archi\
    Projet_Richelieu\elevation.txt');
t=elevation(:,1);
fj=0.01:0.001:2;
wj=2*pi*fj;

RAO_heave_interp = interp1(f, RAO_heave(:,2:end), fj, 'linear');
RAO_pitch_interp = interp1(f, RAO_pitch(:,2:end), fj, 'linear');
RAO_roll_interp = interp1(f, RAO_roll(:,2:end), fj, 'linear');
RAO_surge_interp = interp1(f, RAO_surge(:,2:end), fj, 'linear');
RAO_sway_interp = interp1(f, RAO_sway(:,2:end), fj, 'linear');
RAO_yaw_interp = interp1(f, RAO_yaw(:,2:end), fj, 'linear');

S_heave = zeros(length(wj), 1);
S_pitch = zeros(length(wj), 1);
S_roll = zeros(length(wj), 1);
S_surge = zeros(length(wj), 1);
S_sway = zeros(length(wj), 1);
S_yaw = zeros(length(wj), 1);

dwj=wj(2)-wj(1);

for k = 1:length(wj)
    % RAOs at this frequency (first column = 90  )
    RAO_h = RAO_heave_interp(k, 1);
    RAO_p = RAO_pitch_interp(k, 1);
    RAO_r = RAO_roll_interp(k, 1);
    RAO_sur = RAO_surge_interp(k, 1);
    RAO_sw = RAO_sway_interp(k, 1);
```

```
    RAO_y = RAO_yaw_interp(k, 1);

    % Contribution to the time response
    S_heave(k) = JONSWAP(wj(k)).* RAO_h^2;
    S_pitch(k) = JONSWAP(wj(k)).* RAO_p^2;
    S_roll(k) = JONSWAP(wj(k)).* RAO_r^2;
    S_surge(k) = JONSWAP(wj(k)).* RAO_sur^2;
    S_sway(k) = JONSWAP(wj(k)).* RAO_sw^2;
    S_yaw(k) = JONSWAP(wj(k)).* RAO_y^2;
end

% We use the trapezoidal method to integrate
wj = wj(:);
% Calculation of zero-order moments
m0_heave = trapz(wj, S_heave);
m0_pitch = trapz(wj, S_pitch);
m0_roll = trapz(wj, S_roll);

% Calculation of second-order moments
m2_heave = trapz(wj, (wj.^2) .* S_heave);
m2_pitch = trapz(wj, (wj.^2) .* S_pitch);
m2_roll = trapz(wj, (wj.^2) .* S_roll);

% Ship significant response
xr_heave = 4 * sqrt(m0_heave);
xr_pitch = 4 * sqrt(m0_pitch);
xr_roll = 4 * sqrt(m0_roll);

% Zero up-crossing period
Tz_heave = 2 * pi * sqrt(m0_heave / m2_heave);
Tz_pitch = 2 * pi * sqrt(m0_pitch / m2_pitch);
Tz_roll = 2 * pi * sqrt(m0_roll / m2_roll);

% Most probable maximum amplitude
T = 3 * 3600;
xmpm_heave = sqrt(2 * m0_heave * log(T / Tz_heave));
xmpm_pitch = sqrt(2 * m0_pitch * log(T / Tz_pitch));
xmpm_roll = sqrt(2 * m0_roll * log(T / Tz_roll));

fprintf('\n-- Ship significant response (xr) --\n');
fprintf('Heave : %.4f m\n', xr_heave);
fprintf('Pitch : %.4f deg (%.4f rad)\n', xr_pitch, xr_pitch*180/pi);
fprintf('Roll  : %.4f deg (%.4f rad)\n', xr_roll, xr_roll*180/pi);

fprintf('\n-- Zero up-crossing period (Tz) --\n');
fprintf('Heave : %.2f s\n', Tz_heave);
fprintf('Pitch : %.2f s\n', Tz_pitch);
fprintf('Roll  : %.2f s\n', Tz_roll);

fprintf('\n-- Most probable maximum amplitude (T = %.1f h) --\n', T
    /3600);
fprintf('Heave : %.4f m\n', xmpm_heave);
fprintf('Pitch : %.4f deg = %.4f rad\n', xmpm_pitch, xmpm_pitch*180/pi);
fprintf('Roll  : %.4f deg = %.4f rad\n', xmpm_roll, xmpm_roll*180/pi);
```

Listing 4: Spectrum : 90°

This algorithm provides spectral analysis of ship motions in beam seas (90° waves). It calculates response spectrum by combining JONSWAP wave spectrum with ship RAOs, then computes main statistical parameters using spectral moments : significant response amplitudes, zero up-crossing periods, and most probable maximum amplitudes over 3 hours (*see Sections 4.2 and 4.3*).